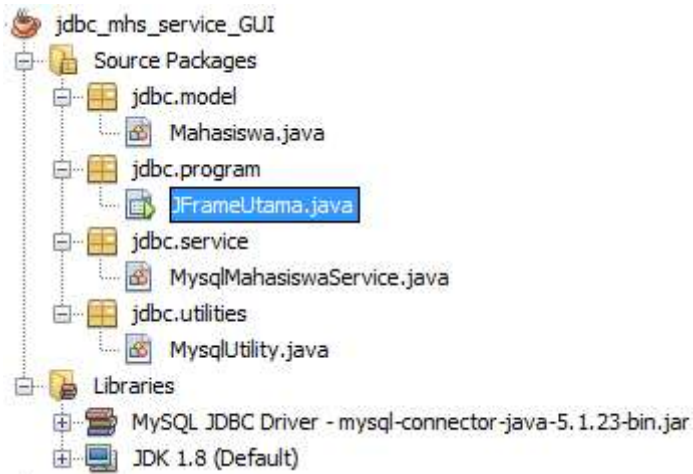


Praktikum 10

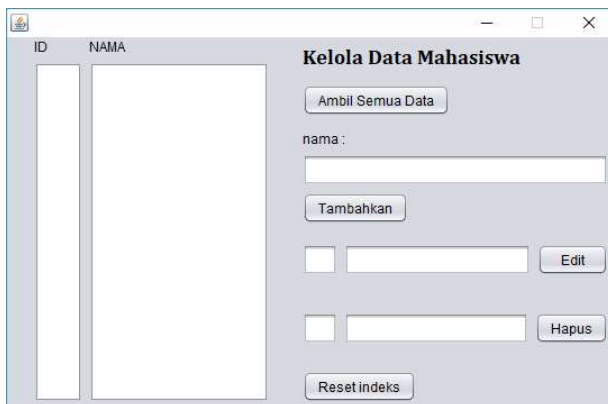
Akses Basis Data

3.2. Menambahkan GUI sebagai antarmuka pengguna

- Salin project `jdbc_mhs_service`, beri nama baru `jdbc_mhs_service_GUI`
- Hapus main class yang lama, gantikan dengan JForm Sebagai main Class



- Rancang GUI yang sesuai untuk mendemonstrasikan aktifitas CRUD



Tambahkan beberapa method pada kelas Service jika dibutuhkan, misalnya sebagai berikut:

Praktikum 10

Akses Basis Data

```
// Constructor
public MysqlMahasiswaService() {...3 lines }

/** Membuat objek mahasiswa ...4 lines */
public Mahasiswa makeMhsObject() {...3 lines }

/** Menambahkan data mahasiswa ...3 lines */
public void add(Mahasiswa mhs) {...23 lines }

/** Update data mahasiswa ...4 lines */
public void update(Mahasiswa mhs) {...22 lines }

/** Delete data mahasiswa sesuai id ...5 lines */
public void delete(int id) {...22 lines }

/** Ambil mahasiswa sesuai id ...6 lines */
public Mahasiswa getById(int id) {...33 lines }

/** Ambil semua isi tabel mahasiswa ...5 lines */
public List<Mahasiswa> getAll() {...39 lines }

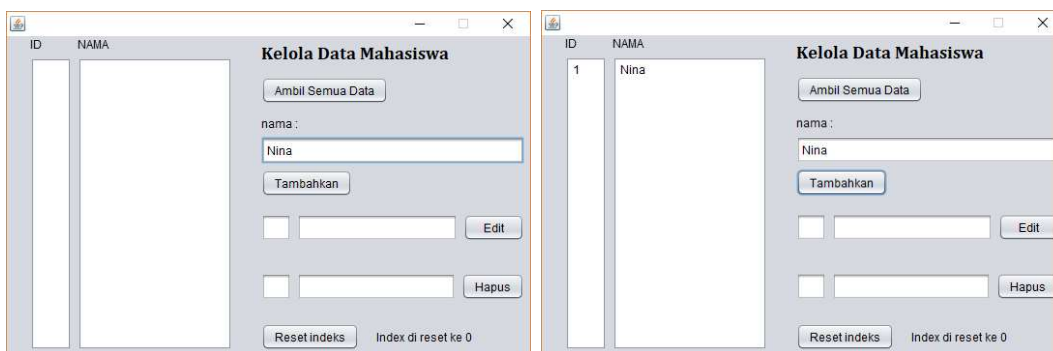
/** Reset indeks tabel mahasiswa ke 1 ...3 lines */
public void indexReset() {...21 lines }

/** Periksa apakah tabel kosong ...5 lines */
public boolean isEmpty() {...23 lines }

/** Menutup koneksi ke DB ...3 lines */
public void closeConnection() {...8 lines }
```

d. Create (insert/add)

Tambahkan **Nina** ke tabel mahasiswa, kemudian tampilkan isi tabel pada GUI



Periksa isi basis data untuk memastikan bahwa Nina berhasil ditambahkan:

```
mysql> select * from jdbc_mhs.mahasiswa;
+-----+
| id | nama |
+-----+
| 1 | Nina |
+-----+
```

Praktikum 10

Akses Basis Data

e. Read (selectAll/getAll)

Tambahkan Rudi dan Beni

The screenshot shows a web application titled "Kelola Data Mahasiswa". On the left, there is a table with two columns: "ID" and "NAMA". The table contains three rows: ID 1 with name "Nina", ID 2 with name "Rudi", and ID 3 with name "Beni". To the right of the table, there are several controls: a button "Ambil Semua Data", a text input field labeled "nama:" containing "Beni", a "Tambahkan" button, an "Edit" button, a "Hapus" button, and a "Reset indeks" button with the text "Index di reset ke 0".

Periksa basis data:

```
mysql> select * from jdbc_mhs.mahasiswa;
+----+-----+
| id | nama |
+----+-----+
| 1  | Nina |
| 2  | Rudi |
| 3  | Beni |
+----+-----+
```

f. Update (edit)

Ubah nama Rudi menjadi Agus

The three screenshots show the application at different stages of the update process. In the first screenshot, the "Edit" button is clicked, and the name "Rudi" is entered in the "nama:" field. In the second screenshot, the "Edit" button is clicked again, and the name "Agus" is entered in the "nama:" field. In the third screenshot, the "Edit" button is clicked again, and the name "Agus" is entered in the "nama:" field. The table on the left remains unchanged in all three screenshots.

Periksa basis data:

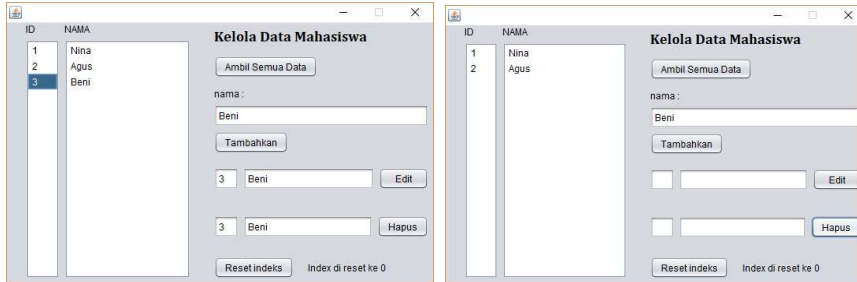
```
mysql> select * from jdbc_mhs.mahasiswa;
+----+-----+
| id | nama |
+----+-----+
| 1  | Nina |
| 2  | Agus |
| 3  | Beni |
+----+-----+
```

Praktikum 10

Akses Basis Data

g. Delete

Hapus Beni dengan cara menyorot ID nomor 3, kemudian klik Hapus



Periksa basis data:

```
mysql> select * from jdbc_mhs.mahasiswa;
+----+-----+
| id | nama |
+----+-----+
| 1  | Nina |
| 2  | Agus |
+----+-----+
```